QUANTUM HACK CHALLENGE · HELL0

What we solved, and how

We're hell0, four bachelor's students at Aalto with no quantum background before this weekend. We solved 41 of the 49 circuits — more than any other team — all on our laptops. Here's the full breakdown and the methods behind it.

Team hell0   **For** QMill · david.karpuk@qmill.com   **Solved** 41 / 49

01 / Results

Every circuit, and what we used

#	CIRCUIT	Q	METHOD	STATUS
VERY EASY				
1	8_1	8	statevector	✓
2	16_2	16	statevector	✓
3	24_3	24	statevector	✓
4	28_4	28	MPS sweep	✓
5	32_5	32	MPS + verify	✓
6	36_6	36	MPS + verify	✓
7	40_7	40	MPS sweep	✓
8	48_8	48	MPS + verify	✓
9	56_9	56	MPS + verify	✓
10	64_10	64	MPS + verify	✓
EASY				
11	8_11	8	statevector	✓
12	16_12	16	statevector	✓
13	24_13	24	statevector	✓

#	CIRCUIT	Q	METHOD	STATUS
14	32_14	32	MPS sweep	✓
15	36_15	36	Pauli backprop	✓
16	40_16	40	MPS sweep	✓
17	40_17	40	MPS sweep	✓
18	40_18	40	MPS marginals	✓
19	48_19	48	MPS marginals	✓
20	48_20	48	MPS + verify	✓
21	48_21	48	MPS marginals	✓
22	56_22	56	MPS + verify	✓
23	56_23	56	MPS marginals	✓
24	56_24	56	MPS marginals	✓
25	64_25	64	permuted MPS	✓
26	64_26	64	permuted MPS	✓
MODERATE				
27	8_27	8	statevector	✓
28	16_28	16	statevector	✓
29	24_29	24	MPS	✓
30	32_30	32	MPS	✓
31	48_31	48	MPS	✓
32	48_32	48	MPS	✓
33	56_33	56	MPS	✓
34	64_34	64	MPS	✓
HARD				
35	40_35	40	Pauli / MPS	✓
36	48_36	48	Pauli / MPS	✓
37	48_37	48	Pauli / MPS	✓
38	56_38 ★	56	Pauli backprop	✓

#	CIRCUIT	Q	METHOD	STATUS
39	56_39	56	Pauli backprop	✓
40	64_40	64	Pauli / MPS	✓
41	64_41	64	Pauli backprop	✓
OPEN — UNSOLVED BY EVERY TEAM				
42	48_42	48	MPO cancellation (unfinished)	open
43	56_43	56	MPO cancellation (unfinished)	open
44	64_44	64	—	open
45	72_45	72	—	open
46	80_46	80	—	open
47	88_47	88	—	open
48	96_48	96	—	open
49	104_49	104	—	open

★ `56_38` we believe no other team cracked.

`02 / Methods`

The toolkit

We matched a method to each circuit rather than leaning on one, and started every circuit with a cheap structural triage that told us which tool was worth running.

TRIAGE

Odd-parity defect count. Count the qubit pairs touched by an odd number of CX gates — the entangling links the inverse-pair structure can't cancel. It reads off the gate list in milliseconds and predicts a circuit's real difficulty far better than its qubit or gate count, so we never spent compute on the wrong method. `defect_count.py`

WHAT CRACKED THE 41

Exact statevector. Small circuits: build the full state, read the peak directly.

Bond-capped MPS sweeps. Greedy max-marginal decoding through the mid-range; the peak survives truncation because these outputs are so concentrated. `crack.py`

Spectral (Fiedler) MPS reordering. Relabel the qubit line by the CX interaction graph so frequently-coupled qubits sit adjacent, which removes most of the SWAP churn that dominates MPS on dense long-range circuits — this got the 64-qubit ones. `mps_marginal2.py`

$\langle Z_i \rangle$ marginal decoding. Read each qubit's average and threshold it; cheap and parallel, used across several 48–56q circuits.

Find-cheap / verify-exact. A candidate can come from a cheap approximate run, but we certify it with one truncation-free single-amplitude contraction `|⟨s|C|0⟩|²` — above 0.5 proves the peak is unique. This certified a 64-qubit answer on a laptop. `crack.py`

Vectorized Pauli backpropagation. Push each qubit's measurement back through the circuit as a sum of Pauli terms (scaled to ~1M terms), read the bit from the sign. Cracked `56_38`, the hardest circuit; with no exact check possible, rising the term cap until the answer stops changing was our trust criterion. `pauli_fast_b.py`

BUILT FOR THE TAIL (42–49, UNFINISHED)

Light-cone bulk cancellation. Tensor-network simplification cancels the even-parity bulk exactly, so most qubits collapse to a cheap exact contraction. `lightcone_z.py`

Two-stage cancellation. Simplify the cancelled state once and reuse it per qubit, instead of re-simplifying for each one. `fast_z2.py`

Defect-graph reduced density matrix. Compute $\langle Z_i \rangle$ from the small simplified defect cone. We also tested belief propagation here; it diverges on these signed-amplitude networks, so we fell back to exact contraction of the cone. `bp_z.py`

Two-sided MPO cancellation + unswapping. Collapse the inverse-pair bulk and recover the hidden permutation (after Kremer–Dupuis). Validated and converging, but ~hours per 20,000-gate circuit, and we ran out of compute. `pcsim/unswap.py`

RULED OUT, AND WHY

No Clifford shortcut (the angles aren't near π/2 multiples). Qiskit's `O3` removes zero two-qubit gates. Commutation-aware cancellation won't telescope through the rewrites. Deleting even-parity CX is confidently wrong. Each dead end told us where the real difficulty sat.

THE GRADER

We reverse-engineered the submission API: it scores by sampled XEB (not the exact match the rules describe), a wrong guess leaks nothing, and the reward is $\texttt{qubits} \times 1.5^{\texttt{difficulty}}$. We used it as a yes/no oracle to pin the last uncertain bit of `56_38`.

The two ideas we'd reuse anywhere are the defect count (a quick classical read on whether an obfuscated circuit is genuinely hard) and the find-cheap / verify-exact split

— both work on any peaked circuit, and the even-parity fingerprint belongs to any obfuscation built from inverse pairs and rotations.

Team hell0 · Quantum Hack Challenge · for QMill

Gharibyan et al., *Heuristic Quantum Advantage with Peaked Circuits* (2025), arXiv:2510.25838

Kremer & Dupuis, *Efficient Classical Simulation of Heuristic Peaked Quantum Circuits* (2025), arXiv:2604.21908